PROJECT REPORT

On

Crime Rate Prediction

~ A Machine Learning-Based System for Forecasting and Analyzing Cyber Crime Trends

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Crime Rate Prediction

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A B S T R A C T

Crime is a pervasive issue that threatens global security and societal well-being. With the rapid growth of urban populations and the advent of technology, criminal activities have become more complex and widespread, ranging from traditional offenses like theft and assault to modern cybercrimes. Predicting crime is a challenging yet crucial task for law enforcement agencies to allocate resources effectively and implement preventive measures. Traditional methods of crime analysis often fall short in handling the vast and dynamic nature of criminal data. However, advancements in machine learning offer a promising solution by enabling the analysis of large datasets to identify patterns and predict future criminal activities.

This project aims to develop a Crime Rate Prediction System using machine learning techniques to forecast crime trends and identify high-risk areas. By leveraging historical crime data, including factors such as location, time, and type of crime, the system will apply various machine learning algorithms, such as linear regression, decision trees, and neural networks, to build predictive models. The primary objective is to convert crime data into actionable insights, enabling law enforcement agencies to anticipate criminal activities and devise strategies to reduce crime rates.

The system will focus on analyzing crime patterns, identifying hotspots, and predicting the likelihood of future crimes in specific regions or timeframes. By transforming crime prediction into a regression problem, the project aims to improve the accuracy of crime forecasting and provide a tool for officials to make data-driven decisions. The outcomes of this project have the potential to enhance public safety, optimize resource allocation, and contribute to the development of smarter, more efficient policing strategies.

Keywords:
Crime Prediction
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Models Law
Enforcement Crime
Hotspots

1. Introduction

1.1 Background and Context

Crime is an act that is prohibited by law and is punishable by a fine, imprisonment, or other legal action. Every day, reports of criminal activity fill our news outlets and social media platforms, painting a picture of a world in which crime is an ever-present concern. From robberies and violent assaults to cybercrimes and white-collar fraud, there is seemingly no end to the number of ways in which criminals can cause harm. Crime has been a part of human civilization since time immemorial. It has become increasingly prominent in today's world. The rise of technology has created a variety of new crimes, while the emergence of globalization has made the world a smaller place, allowing criminals to move and operate in different countries.

Crime is uncertain and cannot be predicted. Crime prediction is significant to determine increase or decrease in crime rate from preceding years. A huge number of crimes happen every second in different places, in different patterns and in different times and the number is increasing each growing day. A good prediction technique provides a more rapid evolution of criminal data sets. It helps in predicting the correct place of crime and criminal activity, as well as aids in keeping track of resources pertaining to the analysis of crime.

Crime prediction using machine learning is an emerging field of study that uses sophisticated algorithms and data-driven methods to detect and predict criminal activities. Machine learning algorithms can be used to identify patterns in data that may indicate a future crime, such as past criminal activities, demographic information, and environmental factors. By leveraging such data, machine learning can be used to create predictive models that identify the likelihood of a certain crime occurring in a particular area or time frame. Additionally, machine learning can be used to develop insights into the behavior of criminals, helping law enforcement professionals better understand and address criminal activity.

1.2 Problem Statement

Crime is a major problem in today's world, and it is a threat to global security. The population of cities is constantly increasing, resulting in an increase in crime rates. Officials are tasked with the monumental challenge of accurately predicting future crime rates and attempting to reduce them. To help in this regard, various large datasets have been reviewed, extracting information such as location and crime type. Crime prediction utilizes various methods to identify areas that are likely to experience higher levels of crime. Given a set of historical crime data, develop a predictive model to identify areas of high risk for future criminal activity, to improve the accuracy of crime prevention and policing efforts. These methods include analyzing past crime data, identifying crime hotspots, and utilizing predictive analytics.

1.3 Objectives & Scope of the Project

The main aim of this project is to develop a system that can accurately predict crime rates and identify potential future crime trends. This information can then be used by officials to devise strategies to reduce crime rates and create a safer environment. To predict the crime rate (dependent variable) based on the year, location, and type of crime (independent variables), various types of machine learning algorithms will be applied. The system will examine how to convert the crime information into a regression problem, thus helping the officials to solve crimes faster. Crime analysis using available information to extract patterns of crime. Based on the territorial distribution of existing data and the recognition of crimes, various multi-linear regression techniques can be used to predict the frequency of crimes.

Scope:

1. Data Collection and Preprocessing:

    - Data Sources: Collect historical crime data from police records, government databases, public reports, and open data platforms.
    - Data Types: Include structured data (e.g., crime type, location, time, date) and unstructured data (e.g., crime reports, witness statements).
    - Data Cleaning: Handle missing values, outliers, and inconsistencies in the dataset.
    - Data Normalization: Normalize numerical data and encode categorical variables for machine learning models.
    - Feature Engineering: Create new features (e.g., time of day, day of the week, season) to improve model performance.

2. Exploratory Data Analysis (EDA):

    - Statistical Analysis: Analyze the distribution of crime data, including frequency, trends, and correlations.
    - Insight Generation: Identify key factors influencing crime rates, such as population density, socioeconomic conditions, and weather.

3. Model Development and Training:

   - Algorithm Selection: Test and compare various machine learning algorithms, including linear regression, decision trees, random forests, support vector machines (SVM), and neural networks.

   - Model Training: Train models using historical crime data and evaluate their performance using metrics like accuracy, precision, recall, and F1-score.

   - Hyperparameter Tuning: Optimize model parameters to improve prediction accuracy and reduce overfitting.

   - Validation: Use cross-validation techniques to ensure the model generalizes well to unseen data.

4. Crime Hotspot Identification:

   - Geospatial Analysis: Use geographic information systems (GIS) to map crime hotspots and identify high-risk areas.

   - Clustering Techniques: Apply clustering algorithms (e.g., k-means, DBSCAN) to group similar crime incidents and detect patterns.

   - Interactive Maps: Develop interactive maps to visualize crime predictions and hotspots for law enforcement agencies.

5. System Development and Deployment:

   - User Interface: Design a user-friendly dashboard or web application for law enforcement agencies to access crime predictions and insights.

   - Scalability: Ensure the system can handle large volumes of data and scale as needed.

   - Cloud Deployment: Deploy the system on cloud platforms (e.g., AWS, Google Cloud) for accessibility and real-time processing.

   - Security: Implement robust security measures to protect sensitive crime data.

6. Real-Time Data Integration:

   - Real-Time Feeds: Integrate real-time data sources such as social media, surveillance systems, and IoT devices to provide dynamic crime predictions.

   - Alert Systems: Develop real-time alert systems to notify law enforcement agencies of potential criminal activities.

   - Streaming Data Processing: Use streaming data processing frameworks (e.g., Apache Kafka, Apache Flink) to handle real-time data.

7. Collaboration and Feedback:

   - Stakeholder Engagement: Collaborate with law enforcement agencies, policymakers, and researchers to ensure the system meets practical needs.

   - User Feedback: Gather feedback from users to improve the system's functionality,
   - usability, and accuracy.

   - Pilot Testing: Conduct pilot tests in selected regions to evaluate the system's
   - effectiveness and make necessary adjustments.

8. Advanced Techniques and Future Enhancements:

   - Deep Learning: Explore advanced deep learning models (e.g., LSTM, CNN) for time-series crime prediction and pattern recognition.

   - Natural Language Processing (NLP): Use NLP techniques to analyze unstructured data such as crime reports and social media posts.

   - Multimodal Data Integration: Incorporate additional data sources (e.g., weather data, socioeconomic factors, traffic patterns) to enhance prediction accuracy.

   - Expanded Crime Coverage: Extend the system to predict a wider range of crimes, including cybercrime, white-collar crime, and organized crime.

2 Literature Survey

Various applications of crime rate prediction have been carried out, some of which are listed below:

Prediction of Crime Rate in Banjarmasin City Using RNN-GRU Model proposed by Muhammad Alkaff describes a model to predict the crime rate by using the Recurrent Neural Network (RNN) with the Gated Recurrent Unit (GRU) architecture. The model takes into consideration the inflation rate and discretionary income. GRU is a modified RNN algorithm that is simpler than the Long-Short Term Memory (LSTM) Neural Network and is more effective in adapting to different timescales and dealing with Vanishing Gradient problems. It consists of two gates, the Update gate (zt) and the Reset gate (rt), and is compatible with data that is not as much as LSTM, achieving optimal results even with fewer data. After collecting and normalizing the data, the model produced the best results with the lowest MAE and RMSE values of 1.7368 and 2.21, respectively, and an R-Squared value of 0.84, indicating good model performance.[1]

Empirical Analysis for Crime Prediction and Forecasting Using Machine Learning and Deep Learning Techniques proposed by Wajiha Safat aims to analyze crime prediction in the Chicago and Los Angeles datasets by improving the predictive accuracy with the Logistic Regression, SVM, Naïve Bayes, KNN, Decision Tree, MLP, Random Forest, and XGBoost algorithms, time-series analysis with LSTM, exploratory data analysis for visual summary, and crime forecasting for the crime rate and high-intensity crime areas for subsequent years with an ARIMA model. This paper investigated the predictive accuracy of eight different algorithms for the Chicago and Los Angeles datasets, with XGBoost performing best with an accuracy of 94% and 88%, respectively.To measure scale-dependent error, an LSTM model was implemented, and RMSE and MAE metrics were used. In addition, an ARIMA model was used to forecast future crime density areas, indicating that Chicago will continue to increase moderately, followed by a stable decline, while Los Angeles will decline sharply.[2]

Sakib Mahmud and Musfika Nuha proposed the relationship between crime and different features in the criminology literature. To reduce crimes and detect criminal activity, the author used Z-Crime Tools and Advanced ID3 algorithms with data mining technology, K-Means Clustering and deep learning algorithms, random forest and naïve Bayes algorithms, and multi-linear regression. Additionally, the author used Apriori and Naive Bayes algorithms to identify and predict criminal trends and patterns.

For classification, algorithms such as Naive Bayes were used to classify objects into predefined groups and classes. The accuracy of different algorithms is evaluated, with K-nearest neighbour providing the most precise crime rate forecast system. Linear, Naive Bayes and KNN algorithms had accuracy scores of 73.6%, 69.5% and 76.9% respectively.[3]

Gaurav Hajela proposed a clustering-based hotspot identification approach for crime prediction. The study of crime shows that it can be represented with a spatiotemporal pattern across geographical space.

There are many indicators of crime such as urban or census-based indicators, streetlight and daylight, social media-based indicators, population flow indicators, and climate-based indicators.A crime hotspot is an area with a higher concentration of crime than the rest of the area. This paper proposes a crime prediction model for the dataset of San Francisco, which includes crime hotspot identification, dataset preparation, and crime prediction approach. Results show that the best accuracy is obtained when k=4 and when coupled with hotspot identification. The decision tree approach achieved 83.95 % and outperforms Nave Bayes.[4]

Masoomali Fatehkia used Facebook interests to improve predictions of crime rates in urban areas. This study discusses the potential for using data from the Facebook Advertising API to gain insight into the distribution of individual-level processes concerning crime rates across different neighbourhoods. It begins by describing existing theories of carcinogenesis related to factors such as poverty, social disorganization, income inequality, and impulsivity. It then outlines how the API could be used to measure the prevalence of interests among a ZIP code's Facebook population, which can be used to reflect the behavioural and attitudinal features of a population.

The models used only demographic factors, only Facebook interests, or both, and controlled for each city's baseline crime rate and the age composition of the neighbourhood. Results showed that the combination of demographic factors and Facebook interests had the greatest predictive strength for all three crime types, both in-sample (using adjusted R2) and out-of-sample prediction (using MAE).[5]

Crime Rate Prediction using KNN proposed by Ms. Vrushali Pednekar, Ms. Trupti Mahale, Ms. Pratiksha Gadhave, and Prof. Arti Gore discusses about a system that convert crime information into a data-mining problem to help detectives solve crimes faster. It focuses on crime analysis, extracting target datasets, data pre-processing, data mining, and interpretation and using discovered knowledge.

The proposed model of crime analysis and prediction uses a general algorithm which takes raw data of crime from a government repository as input and produces a correlated dimensions model for crime analysis and prediction as output.

It also uses various data mining techniques to predict the frequency of occurring crime based on territorial distribution of existing data. It also involves data cleaning and treating missing values to improve the quality of data for mining. With the proposed system, real-time data can be analyzed to cluster and predict crimes. The methods proposed for crime prediction do not address parameters such as outlier effects during the data mining preprocessing, the quality of training and testing data, or the value of features. [6]

2. Methodology

2.1 Ensemble Learning

Ensemble learning is a type of machine learning technique that combines multiple individual models to produce better predictive performance than could be achieved from any of the individual models alone. It works by building multiple models from the same training data set, then combining the models to make more accurate predictions. Ensemble learning has been shown to be successful in a wide variety of applications from computer vision to natural language processing. It is popular because it can produce better results with less data and is more robust to outliers in the data. Ensemble learning is used in many areas like image recognition, natural language processing, and medical diagnosis.

2.2 Algorithm - Random Forest

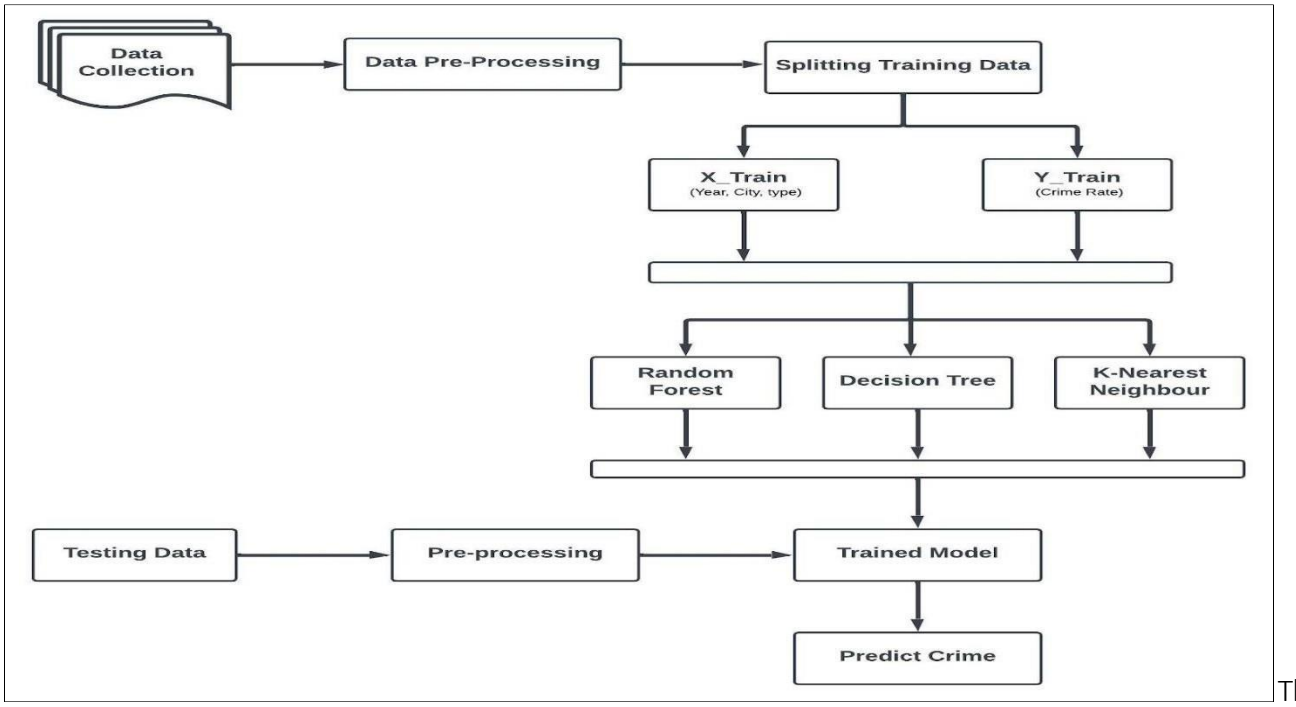
Random Forest is an ensemble learning technique that combines multiple decision trees to generate a more accurate prediction. It is a versatile and powerful algorithm used in a wide variety of applications. The main idea behind the Random Forest algorithm is to combine multiple decision trees into a single model, thereby reducing the variance and improving the accuracy of predictions. This is achieved by randomly sampling data points, randomly selecting features, and then building multiple decision trees on the data. The predictions from each tree are then averaged to produce the final prediction. Random Forests have been shown to be more accurate than traditional decision trees and have become one of the most popular machine learning algorithms. They are also very robust, even when dealing with large datasets, and are resistant to overfitting. The algorithm can be illustrated as follows:

1. Begin with a dataset of observations and their associated labels.
2. Randomly select ‘k’ features from the dataset.
3. For each of the ‘k’ features, choose the best split point.
4. Create a tree using the chosen split points.
5. Repeat steps 2-4 for each of the ‘k’ features.
6. Combine the trees to form a forest.
7. Use the forest to make predictions on new data.
8. Calculate the accuracy of the predictions.

2.3 Proposed System

- Initially the dataset is prepared manually based on the publication available on the National Crime Rate Bureau (NCRB) official website.
- Data Preprocessing: The data is prepared in the correct format for analysis. Some columns are removed or transformed, and label encoding is used to convert the categorical data into numeric for better prediction.
- Random Sampling: After feature selection, the data has been splitted into two parts: training data (70%) and testing data (30%).
- Model Creation: The model algorithms are imported from sklearn. Model is build using model.fit(). The dataset has been analyzed using five different models: support vector machine, nearest neighbor, decision tree, random forest, and neural network.
- Model Selection: Based on the defined goals and model performance, random forest model has been selected. Prediction is done using model.predict(). The model accuracy is determined using accuracy_score imported from metrics.
- Model Deployment: The model has been deployed for prediction using various web technologies.

project has undergone the following process:



The

3. Result

The Random Forest Regression model demonstrates the best accuracy in predicting test data among the five selected models. The model predicts the crime rate value for 10 different categories of crimes, including Murder, Kidnapping, Crime against Women, Crime against Children, Crime Committed by Juveniles, Crime against Senior Citizens, Crime against SC, Crime against ST, Economic Offenses, Cyber Crimes that will occur in 19 Indian metropolitan cities: Ahmedabad, Bengaluru, Chennai, Coimbatore, Delhi, Ghaziabad, Hyderabad, Indore, Jaipur, Kanpur, Kochi , Kolkata, Kozhikode, Lucknow, Mumbai, Nagpur, Patna, Pune, Surat in future.

The accuracy results obtained after testing are listed below:

Algorithm	Mean Absolute Error	Mean Squared Error	R2 Score
Support Vector Machine	10.3204	371.7907	0.17886
K-Nearest Neighbor	6.58181	140.8179	0.55349
Neural Networks MLPRegressor	12.4248	307.5506	0.24823
Decision Tree Regressor	2.89024	34.95932	0.88915
Random Forest Regressor	2.49143	21.43956	0.93201

Fig: Home page

4. Conclusion

Crime rate prediction has become an important tool for law enforcement agencies to help them focus their resources in high-crime areas. With the help of sophisticated algorithms and data analysis, law enforcement agencies can predict when and where crimes are likely to occur. By focusing their resources in the right areas, police officers can help reduce the overall crime rate in a community. Predictive policing has already proven to be an effective tool in reducing crime rates in many areas, and it looks like it will continue to be a key tool in the future.

As a result of machine learning technology, finding relationships and patterns between various data has become easier. The project focuses primarily on predicting the crime rate given the year, city, and types of crime in the future. The training data has bee cleaned and transformed to create a machine learning model using the concept of machine learning. The model predicts the crime rate with an accuracy of 93.20%. The model prediction of crime rate and data visualization helps in analysis of data set and prediction of crimes. Many graphs are created to found interesting statistics that helped in understanding different crime datasets that can be used in implementing the factors that can help in keeping society safe.

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